Sixth-order global reality and general even-order windows for arithmetic theta transforms

Project Stardust

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Abstract

We give a human proof of eventual reality and simplicity of every zero of the cosine transform obtained by multiplying the arithmetic theta kernel by $\exp(tu^6)$. The conclusion holds on the entire complex plane, with one sufficiently late time valid for all frequencies. The proof constructs actual stationary points in a moving transition region and its far complement, proves a uniform oscillatory Gaussian estimate for the full kernel, and converts an analytic cosine-pair approximation into exact real and simple zeros. The sixth-order polynomial changes, physical-prefix bound, and coverage by a common real stationary center are explicit. For every fixed integer-even order we additionally prove real and simple zeros inside an expanding saddle window, with full actual weighted-moment estimates and the precise non-global quantifier boundary. The results are matched to a frozen Lean development. No general integer-order classification, optimal threshold or Riemann-hypothesis consequence is asserted; novelty remains unresolved.

1 Statement and the actual integral

For $u \geq 0$ let

$$K(u) = \sum_{n\geq 1} (2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u}) e^{-\pi n^2 e^{4u}}, \qquad F_t(z) = \int_0^\infty e^{tu^6} K(u) \cos(zu)\, du. \tag{1}$$

The sign of the tilt is positive. Here and below t is real and z is complex. A zero is simple when $F_t'(z) \neq 0$.

Theorem 1.1. *There is a finite T such that, for every $t \geq T$ and every $z \in \mathbb{C}$, $F_t(z) = 0$ implies*

$$\operatorname{Im} z = 0, \qquad F_t'(z) \neq 0.$$

The time T does not depend on z. In particular, a fixed-frequency asymptotic cannot by itself prove this statement. We use one escaping stationary center to cover all frequencies. The proof follows the actual-saddle organization of the quartic companion [1], but supplies the sixth-order derivatives and estimates needed for the extension. It does not import the quartic arithmetic certificate or its endpoint collision.

Each summand in (1) is positive on the half-line. Summation after splitting off half the exponential gives

$$0 < K(u) \leq C e^{9u - \frac{\pi}{2} e^{4u}}, \qquad u \geq 0. \tag{2}$$

Indeed $n^2 e^{4u} \geq (n^2 + e^{4u})/2$ and $\sum n^4 e^{-\pi n^2/2} < \infty$. Thus every $u^k e^{tu^6 + Bu} K(u)$ is integrable, for fixed k, t, B. On compact parameter sets these are integrable dominating functions for all spatial

derivatives. Consequently F_t is entire, even, conjugation compatible, and nonzero since $F_t(0) > 0$. All contour integrals below are ordinary absolutely convergent integrals when their parameters are fixed.

The exact evolution sign

The growing tilt must be distinguished from a stable higher-heat evolution. For any fixed integer $m \geq 1$ define

$$F_{m,t}(z) = \int_0^\infty e^{tu^{2m}} K(u) \cos(zu)\, du.$$

For every nonnegative integer k, differentiation under the integral gives

$$\partial_t \partial_z^k F_{m,t}(z) = (-1)^m \partial_z^{k+2m} F_{m,t}(z). \tag{3}$$

To justify both differentiations, fix a compact time interval and a compact set of z. Choose $A \geq |t| + 1$ and $B \geq |\operatorname{Im} z|$ uniformly there. Every required derivative is bounded in absolute value by a constant times

$$(1 + u^{k+2m}) \exp(Au^{2m} + Bu + 9u - \tfrac{\pi}{2} e^{4u}), \qquad u \geq 0.$$

This is integrable: the double exponential dominates every displayed fixed polynomial and linear exponent at infinity, and the function is bounded on compact intervals. The same bound controls difference quotients locally through their integral derivative formula. Dominated differentiation therefore applies. Finally $\partial_z^{2m} \cos(zu) = (-1)^m u^{2m} \cos(zu)$, which proves (3), including all mixed derivatives. The constants are for a fixed order and fixed derivative count.

In particular the actual transform in Theorem 1.1 satisfies $\partial_t F_t = -\partial_z^6 F_t$. Under the angular Fourier convention e^{iuz}, a stable order-$2m$ multiplier is $e^{-su^{2m}}$, whose spatial generator is $(-1)^{m+1}\partial_z^{2m}$. Its sign is opposite to (3). Within the actual integral, replacing t by $t - s$ inserts precisely that decaying factor; increasing t inserts the growing factor instead. This is an identity of the stated transforms, not a claim about a common completed operator domain. Consequently contraction, positivity or stochastic interpretation cannot be transferred to the increasing-t deformation from the stable multiplier merely by naming the differential order.

The full source `RiemannEvenEvolution` proves the actual cosine identity, mixed jet differentiation and `even_order_heat_equation`; its `sixth_order_heat_equation` is the displayed sixth-order specialization. This elementary evolution identity makes no new priority claim and supplies no reality propagation in the time direction omitted by a higher-order zero theorem.

2 The theta strip and the sixth-order slope

The kernel has an even holomorphic continuation to

$$\mathcal{U} = \{w : |\operatorname{Im} w| < \pi/8\}.$$

Here is the precise kernel information we shall use. Write $w = a + ib$, $E = e^{4w}$, $L = e^{4a}$ and $V = \operatorname{Re} E$. For $a \geq 0$, $V \geq 1$, normal convergence of the series gives

$$K(w) = 2\pi^2 e^{9w - \pi E}(1 + \rho(w)), \qquad |\rho^{(j)}(w)| \leq C_j\{e^{-4a} + (1 + L)^j e^{-3\pi V}\}, \quad 0 \leq j \leq 4. \tag{4}$$

For completeness, after dividing by the leading exponential the remaining factor is

$$1-\frac{3}{2\pi}e^{-4w}+\sum_{n\ge 2}\left(n^4-\frac{3n^2}{2\pi}e^{-4w}\right)e^{-\pi(n^2-1)e^{4w}}.$$

Each of its first four derivatives is bounded by a constant times $(1+L)^j$ times a polynomial in n and the same exponential. For $n\ge 2$, $n^2-1\ge 3$; splitting off this minimum and summing the remaining Gaussian proves (4).

The global strip bound needed at the physical endpoint is

$$|K(s+ib)|\le C\cos(4b)^{-3}e^{-s},\qquad s\ge 0. \tag{5}$$

To check it, compare $\sum n^m e^{-\pi V n^2}$, $m=2,4$, with the Gaussian integral from zero, obtaining $C_m V^{-(m+1)/2}$. Substitution of $V=e^{4s}\cos(4b)$ in the two terms of the kernel gives respectively $Ce^{-s}\cos(4b)^{-5/2}$ and $Ce^{-s}\cos(4b)^{-3/2}$, both bounded as stated.

Evenness follows from the Gaussian theta identity. If $\theta(v)=\sum_{n\in\mathbb{Z}}e^{-\pi n^2 v}$, its modular identity is $\theta(v)=v^{-1/2}\theta(1/v)$, with the right-half-plane square root; see [4]. Put $G(w)=e^w(\theta(e^{4w})-1)/2$. Then $G(-w)=G(w)+\sinh w$, and direct differentiation gives $(D^2-1)G=8K$. This proves evenness throughout $\mathcal{U}$.

For large a, under $V\ge c_0 t a^5$, $t\ge 1$, the kernel is nonzero and its full logarithmic derivative is

$$\sigma_t(w)=6tw^5+\frac{K'(w)}{K(w)}=6tw^5+9-4\pi e^{4w}+\epsilon(w),\qquad |\epsilon^{(j)}(w)|\le Ce^{-4a}\quad(0\le j\le 3). \tag{6}$$

This follows by differentiating $\log(1+\rho)$ in (4). Indeed $L^4e^{-cV}\le e^{16a-cta^5}$ is eventually smaller than e^{-4a}. The same bound holds along $w+s$, $s\ge -1$, with Ce^{-4a} on the right: the exponential terms are bounded by $\exp(16a+16s-cta^5e^{4s})$, whose maximum on this half-line is at its left endpoint for large a. All constants below are independent of t and frequency, unless a fixed parameter is indicated by a subscript.

3 Uniform analysis at an actual saddle

Fix $c_0=10^{-3}$. Call (t,z,v) admissible when

$$t\ge 1,\quad v=a+ib\in\mathcal{U},\quad a\text{ large},\quad y=\operatorname{Im}z\le 1,\quad \sigma_t(v)+iz=0,\quad V\ge c_0 t a^5. \tag{7}$$

These conditions will be proved in the constructions below; no theta asymptotic is applied at an assumed model root. Set

$$Y=\max(0,-y),\quad Q=ta^5+Y,\quad \beta=-\sigma_t'(v)/2,\quad W_z(w)=e^{tw^6+izw}K(w).$$

Taking real parts in full stationarity gives

$$4\pi V=6t(a^5-10a^3b^2+5ab^4)+9-y+\operatorname{Re}\epsilon(v). \tag{8}$$

Since $|b|<\pi/8$, this implies, uniformly,

$$cQ\le V\le CQ,\quad Q\le CL,\quad Q\ge a^5,\qquad cL\le|\beta|\le CL,\quad cQ\le\operatorname{Re}\beta\le CQ. \tag{9}$$

In fact

$$\sigma_t'(v)=30tv^4-16\pi e^{4v}+\epsilon'(v),\qquad \sigma_t''(v)=120tv^3-64\pi e^{4v}+\epsilon''(v).$$

The polynomial terms have size at most $Cta^4 = o(Q)$, whereas the exponential has modulus L and real part V. On any sufficiently small fixed horizontal interval the second derivative is bounded by CL. On the entire horizontal ray $s \geq -1$ one has

$$\operatorname{Re} \sigma_t'(v+s) \leq -cQ. \tag{10}$$

To see this, use $(a+s)^4 \leq Ca^4 e^{2s}$ for $s \geq -1$ and large a. Its contribution $Cta^4 e^{2s}$ is absorbed by $16\pi V e^{4s}$; their ratio is at most C/a on this ray. The error from (6) is negligible.

Proposition 3.1. *There is $\eta(a) \to 0$ such that, uniformly at admissible saddles,*

$$I_z(b) := \int_0^\infty W_z(s+ib)\, ds = W_z(v)\frac{\sqrt{\pi}}{\sqrt{\beta}}(1+\delta), \qquad |\delta| \leq \eta(a), \tag{11}$$

$$\int_0^\infty |W_z(s+ib)|\, ds \leq C|W_z(v)|/\sqrt{Q}. \tag{12}$$

The square root in (11) *has positive real part.*

Proof. On $s \geq -1$ put $P(s) = \int_0^s (\sigma_t(v+r) + iz)\, dr$. The exact scalar differential equation gives $W_z(v+s) = W_z(v)e^{P(s)}$. Full stationarity and (10) imply

$$\operatorname{Re} P(s) \leq -cQs^2, \qquad P(s) = -\beta s^2 + O(L|s|^3) \quad (|s| \leq \rho),$$

where $\rho > 0$ is fixed and small. In particular this argument does not assume a fixed complex disk as wide as the horizontal contour when b is near the strip boundary.

Let $r = L^{-7/16}$. The integral on $[-r, r]$ differs from the Gaussian integral there by $O(Lr^4)$, since $Lr^3 \to 0$ and $|e^h - 1| \leq e^{|h|}|h|$. For $\operatorname{Re}\beta > 0$,

$$\int_{\mathbb{R}} e^{-\beta s^2}\, ds = \sqrt{\pi}/\sqrt{\beta}, \qquad \left|\int_r^\infty e^{-\beta s^2}\, ds\right| + \left|\int_{-\infty}^{-r} e^{-\beta s^2}\, ds\right| \leq \frac{2}{|\beta| r}.$$

The identity follows by differentiating in β and integrating $(se^{-\beta s^2})'$: the product of the integral with $\sqrt{\beta}$ is constant and equals $\sqrt{\pi}$ on the positive real axis. Integration by parts using $(e^{-\beta s^2})' = -2\beta s e^{-\beta s^2}$ proves the tail bound. These are oscillatory bounds in $|\beta|$, not bounds that incorrectly compare its real part to its modulus.

On $r \leq |s| \leq \rho$, Taylor's theorem gives $|P'(s)| \geq cL|s|$ and $|P''(s)| \leq CL$. Integration of $(e^P)'/P'$ bounds these annuli by $C/(Lr)$. The local remainder divided by the main Gaussian is $O(L^{3/2}r^4) = O(L^{-1/4})$, and each normalized Gaussian or annular tail is $O(L^{-1/16})$. The remaining intervals $[-1, -\rho]$ and $[\rho, \infty)$ are bounded by Ce^{-cQ}, using the real-part estimate. After normalization their bounds still tend to zero, since $\sqrt{L} = e^{2a}$ and $Q \geq a^5$.

It remains to treat the physical prefix $0 \leq s \leq a-1$. Write $z = X + iy$. Since $\cos(4b) = V/L \geq cQ/L$, (5) and $\operatorname{Re}(s+ib)^6 \leq s^6 + C$ give

$$|W_z(s+ib)| \leq Ce^{12a}Q^{-3}\exp\{t(a-1)^6 + Ct + Y(a-1) - Xb\}. \tag{13}$$

On the other hand, (4) and (8) give at the center

$$|W_z(v)| \geq \exp\{ta^6 - \tfrac{3}{2}ta^5 - Cta^4 + 9a + Y(a - \tfrac{1}{4}) - Xb - Ca\}. \tag{14}$$

Indeed $\operatorname{Re} v^6 = a^6 - 15a^4b^2 + 15a^2b^4 - b^6$ and $\pi V = \frac{3}{2}t \operatorname{Re} v^5 + (9 - y + \operatorname{Re}\epsilon)/4$. The difference of the polynomial exponents is

$$t\{(a-1)^6 - a^6 + \tfrac{3}{2}a^5 + Ca^4 + C\} = -\tfrac{9}{2}ta^5 + O(ta^4) \leq -cta^5.$$

The Y difference is $-3Y/4$. Thus the prefix integral, divided by $|W_z(v)|$, is at most Cae^{-cQ+Ca}. It vanishes even after multiplication by $\sqrt{L}$. This completes (11). Absolute integration of e^{-cQs^2} and the same prefix bound proves (12). One may, for example, bound η by a constant times $L^{-1/4} + L^{-1/16} + (1+a)e^{-ca^5+Ca}$. □

4 Contours, reflection, and exact local zeros

Let $J_z = \int_0^\infty W_z(s)\,ds$. For real h in the theta strip, a rectangle gives

$$J_z = I_z(h) + i\int_0^h W_z(iy)\,dy.$$

At its right edge the modulus tends to zero: the leading term $-\pi e^{4R}\cos(4h)$ dominates $tR^6+O(R^5)$ and all frequency and kernel polynomial terms. For $y = \operatorname{Im} z$ this yields the exact identities

$$F_t(z) = \frac{I_z(h) + \overline{I_{\bar z}(k)}}{2} + \frac{i}{2}\int_k^h W_z(iq)\,dq, \tag{15}$$

$$F_t(z) = \tfrac{1}{2}\{I_z(h) + I_{-z}(-h)\}. \tag{16}$$

Evenness of K and of the sixth power cancels the vertical integrals in the second formula.

For an admissible saddle with $y \le 0$, the reflected integrands satisfy the exact modulus relation

$$|W_{-z}(s-ib)| = e^{2ys}|W_z(s+ib)|.$$

On $s \ge a-1$ use $e^{2ys} \le e^{2y(a-1)}$ and (12). On the prefix use $e^{2ys} \le 1$ and the prefix bound. With $A_z = W_z(v)\sqrt{\pi}/\sqrt{\beta}$,

$$|I_{-z}(-b)/A_z| \le C\sqrt{L/Q}\,e^{2y(a-1)} + o(1). \tag{17}$$

In particular no such zero has $y \le -1$ for large a: $\sqrt{L} = e^{2a}$ cancels $e^{-2(a-1)}$, leaving $C/\sqrt{Q}+o(1)$, whereas $I_z(b)/A_z \to 1$.

Lemma 4.1. *Let f, E, Φ be holomorphic on $|z-x| \le 1$, x real, with conjugation compatibility and E nonvanishing. Suppose, for a large and $D/a \to 0$,*

$$|\Phi'(z) - a| \le D, \qquad |f(z)/E(z) - \cos\Phi(z)| \le \varepsilon\cosh(\operatorname{Im}\Phi(z)), \qquad \varepsilon \to 0.$$

Then every zero with $|z-x| < 1/2$ is real and simple.

Proof. At a zero q, write $\Phi(q) = u + iv$. The identity $|\cos(u+iv)|^2 = \cos^2 u + \sinh^2 v$ gives $\tanh|v| \le \varepsilon$ and $|\cos u| \le 2\varepsilon$. On the disk $|z-q| \le \delta/a$, $\delta = 10^{-3}$, integration of the derivative bound gives $|\Phi(z) - u| \le 2\delta + 2\varepsilon$. Take ε sufficiently small. If $d = \sin u$, then $|d| \ge 1/2$ and $|\sin\Phi(z) - d|$ is bounded by a small absolute constant. Cauchy's estimate on this disk bounds the derivative of $f/E - \cos\Phi$ on its half-disk by $C\varepsilon a/\delta$. Hence there

$$|(f/E)'(z) + ad| \le a|d|/2.$$

Also vertical integration of Φ' from the real axis gives $|\operatorname{Im}\Phi(q)| \ge (a-D)|\operatorname{Im} q|$. Thus $|q - \bar q| \le C\varepsilon/a < \delta/(2a)$. If q were nonreal, q and $\bar q$ would be distinct zeros in the half-disk. Integration of the last derivative bound along their segment gives $|(f/E)(q) - (f/E)(\bar q)| \ge a|d||q-\bar q|/2 > 0$, a contradiction. The same bound shows the derivative at the real zero is nonzero. Multiplication by E preserves simplicity. □

Proposition 4.2. *If a real x has an admissible saddle $v_0 = a + ib$, then, for large a, every zero z with $|z - x| < 1/2$ is real and simple.*

Proof. The theta boundary has distance at least cQ/L from v_0, because $\cos(4b) \geq cQ/L$. This is much larger than $1/L$. For $|z - x| \leq 2$, solve $\sigma_t(v) + iz = 0$ by the contraction

$$v \longmapsto v - \frac{\sigma_t(v) + iz}{\sigma_t'(v_0)}$$

on a disk of radius C/L about v_0. Its derivative has modulus $O(1/L)$, by the CL bound on σ_t'', and its center displacement is $O(1/L)$. It gives an analytic branch with $v(z) = v_0 + O(1/L)$ and $v'(z) = -i/\sigma_t'(v(z)) = O(1/L)$. For $|z - x| \leq 1$, the perturbed saddle remains admissible: first its margin is at least $(c_0/2)ta^5$, then (8) restores a fixed stronger margin for large a. The quantities a, L, Q remain comparable to their central values.

The nonzero analytic function $A(z) = W_z(v(z))\sqrt{\pi}/\sqrt{\beta(z)}$ has logarithmic derivative

$$\frac{A'(z)}{A(z)} = iv(z) - \frac{\beta'(z)}{2\beta(z)} = iv(z) + O(1/L). \tag{18}$$

Full stationarity cancels the derivative of v in the exponential weight. Also $|\beta'| = O(1)$ because $|\sigma_t''| = O(L)$ and $|v'| = O(1/L)$. Choose $P = \log(A/A(x))$ with $P(x) = 0$, put $P^*(z) = \overline{P(\bar{z})}$ and fix a real argument α of $A(x)$. Define

$$E(z) = |A(x)|e^{(P+P^*)/2}, \qquad \Phi(z) = \alpha + (P - P^*)/(2i).$$

Both have the required conjugation property and $\Phi' = a + O(1/L)$. Exactly,

$$\tfrac{1}{2}(A(z) + \overline{A(\bar{z})}) = E(z)\cos\Phi(z), \qquad \tfrac{1}{2}(|A(z)| + |A(\bar{z})|) = |E(z)|\cosh(\operatorname{Im}\Phi(z)).$$

Use (15) with heights $h = \operatorname{Im} v(z)$ and $k = \operatorname{Im} v(\bar{z})$. Proposition 3.1 controls its two horizontal integrals by these A terms. The heights differ by $O(|\operatorname{Im} z|/L)$, and throughout their intervening interval $\cos(4q) \geq cQ/L$. Equations (5) and $\operatorname{Re}(iq)^6 = -q^6 \leq 0$ give a bound $\exp(Ca + t - Xh + C)$ for the residual vertical integral, where $X = x$ may replace $\operatorname{Re} z$ at bounded cost. At the central real saddle, stationarity implies $|x| \leq CL$, so $|X(h - k)| = O(1)$. By (14), meanwhile, $|A(z)| \geq \exp(ta^6 - Cta^5 - Ca - Xh)$. The ratio tends uniformly to zero since $t \geq 1$ and $a \to \infty$. Thus

$$|F_t(z)/E(z) - \cos\Phi(z)| \leq o(1)\cosh(\operatorname{Im}\Phi(z)), \qquad |z - x| \leq 1.$$

Lemma 4.1 applies with $D = C/L$. □

5 The moving transition region

The positive continuous function $e^{tu^6}K(u)$ tends to zero at infinity and attains a maximum on $[0, \infty)$. Every such maximizer $u = u_t$ escapes to infinity as $t \to \infty$. Indeed for a fixed R, the values on $[0, R]$ are at most $C_R e^{tR^6}$, whereas the value at $R+1$ is $K(R+1)e^{t(R+1)^6}$ and eventually larger. At a positive maximizer differentiation gives $\sigma_t(u) = 0$. With $B = 4\pi e^{4u}$,

$$B = 6tu^5 + 9 + \epsilon(u) \asymp tu^5. \tag{19}$$

This is the center used in both frequency regions.

Proposition 5.1. *Fix $M > 0$. For all sufficiently large t, every zero of F_t in $|z| \le MB$ is real and simple.*

Proof. It suffices first to take $\operatorname{Im} z \le 0$. The anchor $q = u + \frac{1}{4}\operatorname{Log}(1 + iz/B)$ has $q - u = O_M(1)$ and a fixed positive distance, depending on M, from the theta boundary. The exact fixed-point equation is

$$w = u + \frac{1}{4}\operatorname{Log}\left(1 + \frac{iz}{B} + \frac{6t(w^5 - u^5) + \epsilon(w) - \epsilon(u)}{B}\right). \tag{20}$$

On the disk $|w - q| \le C_M/u$, the logarithm argument stays in a right half-plane: its leading real part is at least one and its perturbation is $O_M(1/u)$. The derivative of the perturbation is $O_M(tu^4/B) = O_M(1/u)$. At q the displacement is $O_M(1/u)$; taking C_M large and then u large makes the disk invariant and the map contractive. The root is a full actual saddle with

$$\operatorname{Re} v = u + O_M(1), \qquad \operatorname{Re} e^{4v} \asymp_M B, \qquad e^{4\operatorname{Re} v} \asymp_M B.$$

In particular the initial theta margin is positive; the real stationarity equation strengthens it to $\operatorname{Re} e^{4v} \ge cta^5$ with an absolute $c > c_0$ for large u. Also $L/Q \le C_M$.

For $y \le -A_M/u$, the reflection bound (17) is at most $C_M e^{-A_M} + o(1)$, since $a - 1 \ge u/2$. Choose A_M large and then u large to make this less than $1/4$, while the relative error in (11) is less than $1/4$. Equation (16) excludes a zero. Conjugation therefore confines every zero in this region to $|\operatorname{Im} z| < A_M/u$.

For $x = \operatorname{Re} z$, $|x| \le MB$, the same construction supplies an admissible real-frequency saddle. Once u is large enough that $A_M/u < 1/2$, Proposition 4.2 proves the zero real and simple. Every estimate is uniform in this bounded-ratio region. □

6 Actual far saddles and whole-plane coverage

Lemma 6.1. *There is an absolute C_0 such that, if $t \ge 1$, $X = |z|$, $\operatorname{Im} z \le 0$, $\alpha = \frac{1}{4}\log(X/(4\pi))$ is sufficiently large and*

$$X \ge C_0 t(\alpha + 1)^5, \tag{21}$$

then z has an admissible actual saddle with real part $\alpha + O(1)$ and $e^{4a} \asymp X$.

Proof. Start with the leading model, not the actual kernel. The anchor $q = \frac{1}{4}\operatorname{Log}(iz/(4\pi))$ has real part α and imaginary part in $[-\pi/8, \pi/8]$, since $\operatorname{Re}(iz) \ge 0$. Solve

$$w = \tfrac{1}{4}\operatorname{Log}\{(iz + 6tw^5 + 9)/(4\pi)\}$$

on the disk $|w - q| \le Ct(\alpha + 1)^5/X$. Choose C_0 so large that this radius is less than $1/100$. On the disk $|\operatorname{Im} w| < 1/2$, and $\operatorname{Re} w^5 \ge (\operatorname{Re} w)^5/2$ for large α. Hence the logarithm argument lies strictly in the right half-plane. Its modulus is comparable to X, and the map's derivative is $O(t(\alpha+1)^4/X) = O(1/\alpha)$. The center displacement is bounded by $Ct(\alpha+1)^5/X$, using $|\operatorname{Log}(1+h)| \le 2|h|$ for $|h| \le 1/2$. Increasing the fixed radius constant gives an invariant contraction disk. Its fixed point solves

$$6tw^5 + 9 - 4\pi e^{4w} = -iz.$$

If $w = a + ib$ and $Y = -\operatorname{Im} z$, its real part is

$$4\pi \operatorname{Re} e^{4w} = 6t(a^5 - 10a^3b^2 + 5ab^4) + 9 + Y \ge cta^5 + Y.$$

Since $|b| < 1/2$ and this quantity is positive, in fact $|b| < \pi/8$. Also $e^{4a} \asymp X$. Thus the model root lies strictly inside the theta strip with the required positive margin; boundary anchors cause no missing case.

Now restore the kernel error. The model derivative at its root is

$$30tw^4 - 16\pi e^{4w} = -4iz + t(30w^4 - 24w^5) - 36.$$

Its modulus is between cX and CX, by (21) with C_0 large. On a disk of radius $1/X$, the variation is $O(1)$, while ϵ and its first derivative are $O(e^{-4a})$. Subtracting the full slope divided by the central model derivative gives a contraction: its center displacement is $O(e^{-4a}/X)$, and its derivative is $O(1/X)$. The disk remains inside the strip because its boundary distance is at least $c(ta^5 + Y)/X \gg 1/X$. The resulting actual root preserves the positive margin; (8) restores the fixed admissibility constant. This proves the lemma. □

Proposition 6.2. *For an absolute sufficiently large C_1, the conditions $t \geq 1$, α sufficiently large and $|z| \geq C_1 t(\alpha+1)^5$ imply that every zero at z is real and simple. For each finite T, all nonreal zeros at $1 \leq t \leq T$ lie in one finite disk.*

Proof. Take $C_1 \geq 2C_0$. For $\operatorname{Im} z \leq -1$, Lemma 6.1 and (17) exclude a zero. Conjugation excludes $\operatorname{Im} z \geq 1$. If $|\operatorname{Im} z| < 1$, let $x = \operatorname{Re} z$. For large $|z|$, $|x| \geq |z|/2$, and $\alpha_x = \frac{1}{4}\log(|x|/(4\pi)) \leq \alpha$. Thus $|x| \geq C_0 t(\alpha_x + 1)^5$. Lemma 6.1 supplies a saddle at x. The local pair estimates in its proof hold on $|q - x| \leq 1$. At any proposed zero there, the inequality $|\cos\Phi| \leq \varepsilon \cosh \operatorname{Im}\Phi$ gives $|\operatorname{Im}\Phi| \leq 2\varepsilon$. Vertical integration of $\Phi' = a + O(1/L)$ yields $|\operatorname{Im} z| \leq 2\varepsilon/(a - O(1/L)) < 1/2$. Proposition 4.2 now applies. For $1 \leq t \leq T$, the displayed power condition holds uniformly for all sufficiently large $|z|$, since $(\log X)^5/X \to 0$. This proves the last assertion. □

Proof of Theorem 1.1. Use a real maximizing center u and $B = 4\pi e^{4u}$ from (19). If $X = |z| \geq MB$, write $\alpha = u + d$, so $d \geq \frac{1}{4}\log M$. Then

$$\frac{t(\alpha+1)^5}{X} \leq Ce^{-4d}\left(1 + \frac{d+1}{u}\right)^5 \leq C(2+d)^5 e^{-4d}, \qquad u \geq 1.$$

Choose one fixed $M > 1$ so large that the last expression is at most $1/C_1$ for every $d \geq \frac{1}{4}\log M$. For sufficiently late t, the center u and hence α are large, so Proposition 6.2 settles all $X \geq MB$. For this same fixed M, enlarge the time once more so Proposition 5.1 settles all $X \leq MB$. These regions cover the entire complex plane, using the same u and B. This proves the common-time conclusion. □

7 General integer-even orders: an expanding window

The whole-plane theorem above is sixth order. There is also a narrower result for every fixed positive integer m, which does not require the complex far-saddle argument. Set $n = 2m$ and define the actual transforms

$$F_{m,t}(z) = \int_0^\infty f_{m,t}(v)\cos(zv)\,dv, \qquad f_{m,t}(v) = e^{tv^n}K(v), \qquad M_{m,t} = F_{m,t}(0) > 0.$$

The same double-exponential kernel bound proves all required moments integrable, uniformly on compact parameter sets, for every fixed m. Thus these are entire, even, conjugation-compatible functions. Here the index $m = 1$ means a quadratic tilt and $m = 3$ the sixth-order tilt.

Theorem 7.1. *Fix $m \geq 1$. For all sufficiently large t, every maximizer u of $f_{m,t}$ on $[0,\infty)$ has the following property:*

$$|z| \leq u/2, \quad F_{m,t}(z) = 0 \quad \Longrightarrow \quad \operatorname{Im} z = 0, \quad F'_{m,t}(z) \neq 0.$$

Every such maximizing center tends to infinity with t. Consequently, for each fixed $R \geq 0$, all zeros in $|z| \leq R$ are real and simple at sufficiently late times. All constants and late times may depend on m; the last time may additionally depend on R.

We give complete actual-kernel estimates. The proof reorganizes the formal weighted-tail bounds with unspecified positive constants depending only on the fixed order; it does not replace the full kernel by its leading summand or assume a limiting probability law.

7.1 Real centers and an actual normalization bound

The positive continuous $f_{m,t}$ tends to zero at infinity, so attains a maximum $P = f_{m,t}(u)$. Every maximizer escapes: on $[0,R]$ its value is at most $C_R e^{tR^n}$ for $t \geq 0$, while at $R+1$ it is $K(R+1)e^{t(R+1)^n}$, eventually larger. This excludes every center in $[0,R]$ at once.

The real-axis specialization of (4), including its first two derivatives, gives for $v \to \infty$

$$L(v) := -\frac{K'(v)}{K(v)} = 4\pi e^{4v} - 9 + O(e^{-4v}), \qquad L'(v) = 16\pi e^{4v} + O(e^{-4v}).$$

Indeed write $K = 2\pi^2 e^{9v - \pi e^{4v}}(1+\rho)$ and differentiate $\log(1+\rho)$, whose first two derivatives are $O(e^{-4v})$. Choose one large $N \geq 2n$ such that $L > 0$ and

$$T(v) := \frac{L(v)}{nv^{n-1}} > 0, \qquad \frac{T'(v)}{T(v)} = \frac{L'(v)}{L(v)} - \frac{n-1}{v} \geq 3 \quad (v \geq N). \tag{22}$$

This is possible since the expression tends to four. At each sufficiently late maximum $u > N+1$, stationarity gives $t = T(u)$, whence

$$ce^{3u} \leq t \leq Qe^{4u} \tag{23}$$

for fixed $c, Q > 0$. The first bound integrates (22) from N to u; the second follows from $L(u) = O(e^{4u})$ and $nu^{n-1} \geq 1$. In particular $u \leq Ct$ and $e^{2u} \leq Ct$ at late centers. Moreover

$$(\log f_{m,t})'(v) = nv^{n-1}(t - T(v)), \qquad v \geq N.$$

Hence $f_{m,t}$ increases on $[N,u]$ and decreases to its right.

Normalization must use the actual mass $M = M_{m,t}$ rather than just the peak height. On $|v-u| \leq 1$, the kernel derivative estimates and $t = T(u) = O(e^{4u}/u^{n-1})$ give

$$|(\log f_{m,t})''(v)| = |n(n-1)tv^{n-2} - L'(v)| \leq Ce^{4u}.$$

Taylor's theorem at the stationary center implies $f_{m,t}(v) \geq Pe^{-C}$ for $|v-u| \leq e^{-2u}$, after increasing C. This interval lies in the positive half-line. Integrating gives

$$M \geq c_1 e^{-2u} P, \qquad \frac{P}{M} \leq Ce^{2u} \leq Ct. \tag{24}$$

All these estimates hold uniformly over the actual maximizing centers.

7.2 Both tails with a growing exponential weight

For $s \geq 0$, (22) gives $T(u+s) \geq te^{3s}$. As $e^{3s} - 1 \geq 3s$, integration of the log slope yields

$$f_{m,t}(u+s) \leq Pe^{-ts^2}. \tag{25}$$

Here the sharper exponent from integration is $-(3n/2)tu^{n-1}s^2 \leq -ts^2$. For $0 < r \leq 1/2$ and $0 \leq s \leq r$, one has $T(u-s) \leq te^{-3s}$ and $1 - e^{-3s} \geq s$. For the latter inequality, $1 - e^{-3s} - s$ is concave, zero at zero and positive at $1/2$. Therefore

$$\log P - \log f_{m,t}(u-r) \geq nt(u-r)^{n-1} \int_0^r s\, ds \geq tr^2,$$

since $u - r \geq N \geq 2$ and $n \geq 2$. By monotonicity, the same upper envelope Pe^{-tr^2} bounds $f_{m,t}$ on $[N, u-r]$. On $[0, N]$, use $P \geq f_{m,t}(N+1)$ to obtain

$$\int_0^N f_{m,t}(v)\, dv \leq CPe^{-t\Delta}, \qquad \Delta = (N+1)^n - N^n > 0.$$

Combining this with the interval length u and (24) gives the genuine left-mass bound

$$\frac{1}{M} \int_0^{u-r} f_{m,t}(v)\, dv \leq C\{te^{-t\Delta} + t^2 e^{-tr^2}\}. \tag{26}$$

The second term uses $u \leq Ct$; no compact prefix was discarded.

For $B \geq 0$ split the actual centered weighted first moment into $[0, u-r]$, $[u-r, u+r]$, and $u + (r, \infty)$. All these integrals are finite by the original double-exponential kernel envelope. On the left, $(u-v)e^{B(u-v)} \leq ue^{Bu}$. On the middle interval, $|v-u|e^{B|v-u|} \leq re^{Br}$ and its mass is at most M. On the right, (25) and $Bs \leq B^2/(2t) + (t/2)s^2$ give

$$\int_r^\infty se^{Bs} f_{m,t}(u+s)\, ds \leq \frac{P}{t} \exp\{B^2/(2t) - (t/2)r^2\},$$

because $\int_r^\infty se^{-(t/2)s^2} ds = t^{-1}e^{-(t/2)r^2}$. Consequently

$$\begin{aligned} \frac{1}{M} \int_0^\infty |v-u|e^{B|v-u|} f_{m,t}(v)\, dv \leq Cue^{Bu}\{te^{-t\Delta} + t^2 e^{-tr^2}\} \\ + re^{Br} + C\exp\{B^2/(2t) - (t/2)r^2\}. \end{aligned} \tag{27}$$

The constants do not depend on B or r within the stated ranges.

Choose $B = u$ and $r = e^{-u/2}$. Then $ur \to 0$ and $r^2 = e^{-u}$. The lower time bound gives $tr^2 \geq ce^{2u}$, whereas the upper bound controls $t^2 \leq Q^2 e^{8u}$. For large u, $\Delta \geq e^{-u}$, so both exponentials in braces are at most $e^{-ce^{2u}}$ before their factors of t, t^2. The rightmost exponential tends to zero faster than e^{-u}, since $u^2/(2t) - (t/2)e^{-u} \leq u^2/2 - (c/2)e^{2u}$. After multiplying (27) by u, the left and right tails still tend to zero; the middle term is $ue^{-u/2}e^{ue^{-u/2}} \to 0$. We have proved the crucial uniform actual moment conclusion

$$\frac{u}{M} \int_0^\infty |v-u|e^{u|v-u|} f_{m,t}(v)\, dv \longrightarrow 0. \tag{28}$$

It holds as $t \to \infty$ for every maximizing center, because every such center escapes and all preceding bounds are uniform at those centers.

7.3 Cosine approximation and exact real simple zeros

The elementary exponential estimate $|e^w - 1| \le |w|e^{|w|}$ implies, by applying it to the two terms of the cosine,

$$|\cos(zv) - \cos(zu)| \le e^{u|\operatorname{Im} z|}|z|\,|v-u|e^{|z||v-u|}.$$

Integrating and using (28) gives, for every $\eta > 0$, at one sufficiently late time independent of z,

$$\left|\frac{F_{m,t}(z)}{M} - \cos(uz)\right| \le \eta e^{u|\operatorname{Im} z|}, \qquad |z| \le u. \tag{29}$$

This is a growing-frequency estimate, not just compact convergence.

Take $\eta = 10^{-6}$ and $u \ge 2$. If z is a zero in $|z| \le u/2$, write $q = uz = a + ib$. The identity $|\cos(a+ib)|^2 = \cos^2 a + \sinh^2 b$ and (29) give $\sinh|b| \le \eta e^{|b|}$. Thus $1 - e^{-2|b|} \le 2\eta$, whence $|b| \le 2\eta$. Define $g(w) = F_{m,t}(w/u)/M$. On $|w - a| \le 1$,

$$|w/u| \le u/2 + 1/u \le u, \qquad |g(w) - \cos w| \le e\eta.$$

Indeed $|a| \le |q| \le u^2/2$ and $|\operatorname{Im} w| \le 1$. Cauchy's derivative bound on circles of radius $1/2$ gives $|g'(w) + \sin w| \le 2e\eta$ for $|w - a| \le 1/4$. Since $g(q) = 0$, the cosine bound at q and the derivative bound $|\sin w| \le e$ on the unit-height strip imply $|\cos a| \le \eta e^{2\eta} + 2e\eta < 10\eta$. Hence $|\sin a| > 1/2$. On the disk $|w - a| \le 10\eta$,

$$|g'(w) + \sin a| \le 2e\eta + 10e\eta < 1/4 < |\sin a|/2.$$

Integrate g' along any line segment in this disk. Subtracting the constant derivative $-\sin a$ gives

$$|g(w_1) - g(w_2)| \ge \tfrac{1}{2}|\sin a|\,|w_1 - w_2|.$$

Both q and $\bar{q}$ lie in the disk and are zeros by conjugation compatibility. The lower bound forces $q = \bar{q}$. It also gives $g'(q) \ne 0$, and $g'(q) = F'_{m,t}(z)/(uM)$ proves simplicity. This proves the expanding-window assertion. For a fixed R, take a late time such that every center has $u \ge 2R$; the fixed-disk assertion follows with the same zero argument.

7.4 The exact quantifier boundary

For each fixed m, R there is a late time valid on that disk. The argument does not give a time uniform over every R, nor establish global reality for every integer-even order. Any nonreal zeros, if they exist at late times, must leave each fixed disk. This is not an existence assertion about such zeros. The elementary polynomials $p_t(z) = z^2 + t^2$, $t > 0$, have no zeros in $|z| \le t/2$, yet have the nonreal roots $\pm it$. They are a counterexample to inferring global reality from expanding zero-free windows alone, not models of the arithmetic transform. Their root formula and $|it| = t$ prove both statements immediately. The sixth-order whole-plane theorem requires the separate far-frequency work above. No $A(2)$/RH implication or interpolation of reality thresholds is supplied.

8 Exact formal correspondence and publication scope

The frozen mathematical baseline is

`bab688bb6140cd6316c80680095a914ea0d346db.`

The exact final declaration is

`RiemannSixthTransitionSimpleZeros.`
`eventually_all_zeros_real_simple`.

Its namespace prefix is

`OperatorDerivativeRewrite.Foundation`.

The formal transform is

`RiemannEvenFlow.flow3tz`:

the index three means a sixth power, not a third power. The proof combines the bounded-ratio theorem with

`RiemannSixthFarZeroControl.`
`uniform_actual_far_zeros_real_simple`,

both at a single stationary center. The theorem map records exact source lines and hashes. The human proof consolidates explicit formal constants into uniform inequalities; intermediate regrouped lemmas are not claimed as new named formal statements.

De Bruijn [2] and Newman [3] provide the primary historical comparison for real-zero Fourier transforms. Newman's quadratic propagation theorem does not supply the sixth-order conclusion. Targeted priority review remains necessary; novelty is unresolved.

The theorem gives an existential late time. It gives no numerical optimal threshold, no nonreal-zero certificate at a specified early sixth-order time, no assertion that the set of all reality times is an upper ray, and no threshold collision. The whole-plane argument is presented for the verified sixth-order transform only. Section 7 proves the narrower general integer-even expanding-window result. The source chain runs through `RiemannEvenGrowingMoment`, `RiemannEvenGrowingFourier` and `RiemannEvenWindowZeros`; its exact declarations and the classical human proof reorganizations are in the map. Constants and times are order dependent. No global integer-order classification, real-order interpolation or Riemann-hypothesis implication follows.

References

[1] Project Stardust, *An attained eventual reality threshold for a growing quartic arithmetic transform*, reviewed editorial draft, October 29, 2026. Companion manuscript in `manuscripts/quartic-threshold/`.

[2] N. G. de Bruijn, The roots of trigonometric integrals, *Duke Math. J.* **17** (1950), 197–226. doi:10.1215/S0012-7094-50-01720-0.

[3] C. M. Newman, Fourier transforms with only real zeros, *Proc. Amer. Math. Soc.* **61** (1976), 245–251. `https://sites.math.northwestern.edu/~auffing/papers/Newman.pdf`.

[4] NIST Digital Library of Mathematical Functions, equation 20.7.32, modular theta identity. `https://dlmf.nist.gov/20.7.E32`.